🔬 OptiSystem Lab Project Titles (Research-Oriented)

1. **Design and Performance Analysis of Indoor Optical Wireless Communication (OWC) System Using OptiSystem under Multi-User Scenario.**
2. **Visible Light Communication (VLC)-Based Optical Wireless Network with hi data rate.**
3. **Hybrid LiFi–FSO Communication System Design for High-Speed Indoor–Outdoor Integration Using OptiSystem**
4. **Performance Analysis of FSO Link under Fog, Rain, and Dust Attenuation Using OptiSystem**
5. **Design of High-Speed FSO System for 6G Backhaul Using WDM Technique**
6. **Novel Hybrid WDM/TDM Passive Optical Network Architecture with Enhanced QoS: An OptiSystem Modeling Approach**
7. **Investigation of Atmospheric Turbulence Mitigation Techniques in FSO Using Adaptive Power Control**
8. **Performance Evaluation of Mode Division Multiplexing (MDM) Based Optical System Using Laguerre–Gaussian Modes in OptiSystem under Atmospheric Turbulence Conditions**
10. **Design and Simulation of Multi-Channel WDM-FSO Communication System with different modulation scheme.**
11. **128-Channel DWDM System Design Using Optical Amplifiers in OptiSystem**
12. **Secured Free-Space Optical Communication System with Adaptive Modulation and Channel Coding**
13. Next-**Generation 400 Gb/s PON Architecture Simulation Using OptiSystem with Low-Cost DSP Implementation**
14. **Performance Enhancement of WDM Optical Network Using Dynamic Channel Allocation Techniques**
15. **OptiSystem Modeling of Tunable Optical Filters Using Photonic Crystal Structures for Reconfigurable Networks**
13. **Optimization of EDFA Placement and Length in Long-Haul FSO Communication Systems**
14. **Comparative Study of EDFA, Raman, and SOA Amplifiers in WDM Optical Networks Using OptiSystem.**
15. **Design of Hybrid Optical Amplifier (EDFA + Raman) for Ultra-Wideband Optical Networks**
15. **Performance Comparison of NRZ, RZ, QPSK, and OFDM Modulation Techniques in FSO Communication System**
16. **Performance Comparison of DP-QPSK, OFDM, and CSRZ Modulation Techniques in Coherent Optical Systems**

17. **Performance Analysis of Coherent Optical Communication System Using DP-QPSK and 16-QAM Modulation Techniques in OptiSystem for High-Speed Transmission**

18. **Performance Analysis of Polarization Mode Dispersion (PMD) Effects on High-Speed Coherent Optical Communication Systems Using OptiSystem**

19. **Polarization Mode Dispersion (PMD) Effects on 1.5 Tb/s Coherent Optical Communication Systems Using OptiSystem**

20. **Design and Optimization of 1.2 Tbps Super Dense Wavelength Division Multiplexing Systems Utilizing Hybrid Raman-EDFA and Optical Phase Conjugation for 400 km Transmission**

21. **Machine Learning-Based Quality of Transmission (QoT) Prediction in DWDM Optical Networks**

22. **Design and Simulation of Hybrid UWOC–FSO–Fiber Communication System for Smart Coastal Networks**

23. **Design of 10+ Tbps DWDM Radio-over-Fiber (RoF) System for 5G/6G Fronthaul Using OptiSystem**

24. **Novel Hybrid WDM/TDM Passive Optical Network Architecture with Enhanced QoS: An OptiSystem Modeling Approach**

25. **Simulation Investigation of Quantum FSO–Fiber System Using the BB84 QKD Protocol Under Severe Weather Conditions**

26. **Novel Hybrid WDM/TDM Passive Optical Network Architecture with Enhanced QoS.**

27. **Impact of Modulated Optical Solitons on Long-Haul Fiber Optic Communication Systems for Dispersion and Nonlinearity Management".**

28. **Adaptive Modulation and Coding Schemes for FSO Communication under Dynamic Atmospheric Turbulence Conditions Using Real-time Channel Estimation".**

29. **Terahertz Wireless-Optical Integrated Systems for Beyond 5G Communication: A Performance Study on Data Rate and Latency".**

30. **High-Precision Optical Fiber Sensor Network Simulation for Structural Health Monitoring Using OptiSystem**

31. **Mitigation of Four-Wave Mixing and Crosstalk in High-Capacity DWDM Networks using Chirped Fiber Bragg Gratings and Machine Learning-Based Channel Equalization"**

    Rationale: Nonlinear effects like Four-Wave Mixing (FWM) and crosstalk are major impairments in Dense Wavelength Division Multiplexing (DWDM) systems. This proposal suggests a novel combination of passive compensation (Chirped Fiber Bragg Gratings) with active, intelligent equalization using machine learning to enhance signal integrity.

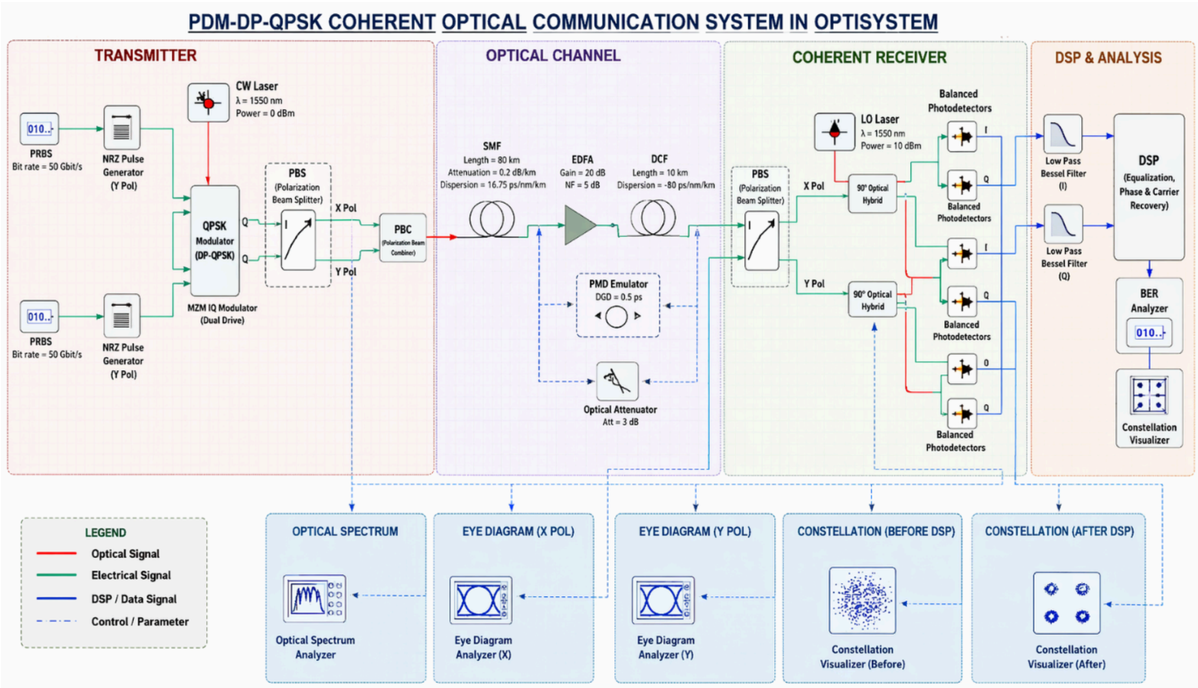
32. **"Design and Analysis of WDM-PON Architectures for Industrial IoT Backhaul Incorporating Secure Chaotic Optical Communication"**

1. Rationale: The Industrial Internet of Things (IIoT) requires robust and secure communication. This integrates WDM-PON for high-bandwidth backhaul with chaotic optical communication for enhanced security, tailored for industrial environments.

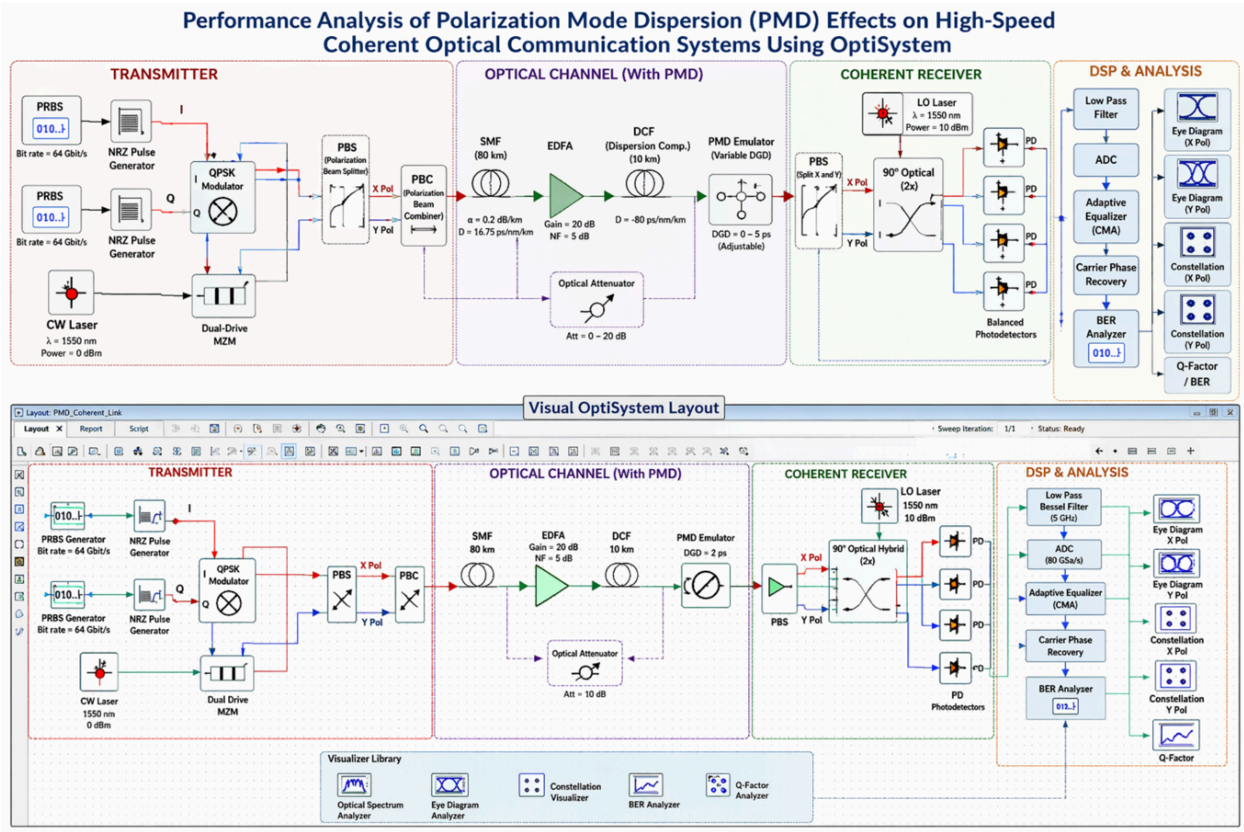
33. **Design and Evaluation of Polarization Division Multiplexing (PDM)-Based Optical Communication System Using DP-QPSK in OptiSystem**

visual OptiSystem layout diagram (exact component placement like software UI) or provide:

Full OptiSystem simulation block diagram and visual OptiSystem layout



34. **“DSP-Based Mitigation of Polarization Mode Dispersion in High-Speed Coherent Optical Systems Using Adaptive Equalization in OptiSystem”**



*Yes—**it is definitely possible to implement (at least at a research/simulation level) in OptiSystem**, but with some **important scope considerations**.*